

DS 6050: Deep Learning Course Project Plan

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Fall 2025

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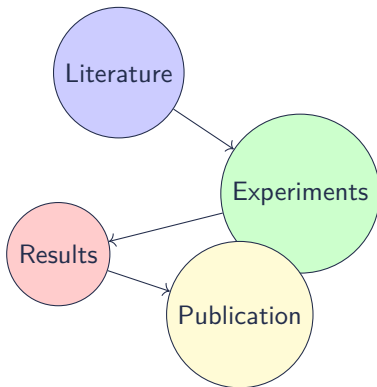
Course Project: Grand Challenge

Objective:

- Apply deep learning concepts to solve real-world problems
- Conduct rigorous research with ablation studies
- Produce publication-quality work

Team Size:

- 2-3 members per team



Critical: Dataset Requirements

Public Dataset Requirement

! Your dataset MUST be publicly available!

Why Public Datasets?

- ✓ Enable paper publication
- ✓ Allow result reproduction
- ✓ Foster academic transparency
- ✓ Build your portfolio

Not Acceptable:

- ✗ NDA-protected data
- ✗ Proprietary datasets
- ✗ Private company data
- ✗ Restricted access data

Recommendation

Choose well-documented benchmark datasets or create your own public dataset with proper documentation.

Recommended Dataset Sources

Computer Vision:

- ImageNet, COCO, CIFAR
- Medical: NIH Chest X-ray, ISIC
- Kaggle Vision Competitions
- Hugging Face Datasets

NLP & Text:

- GLUE, SuperGLUE
- Common Crawl
- WikiText, BookCorpus
- arXiv papers dataset

Time Series & Audio:

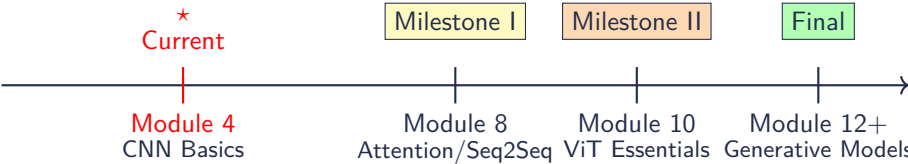
- UCI ML Repository
- Google Speech Commands
- LibriSpeech, AudioSet
- Financial data (Yahoo Finance)

Multimodal:

- MS-COCO Captions
- Visual Question Answering
- CLIP datasets
- Video datasets (UCF101, Kinetics)

Pro Tip: Include dataset URL in your proposal!

Project Timeline & Milestones



Milestone	Due (Module)	Weight
I: Literature Review & Proposal	Module 8	5%
II: Architecture & Preliminary Results	Module 10	10%
III: Final Report & Presentation	After Module 12	25%

Milestone I: Literature Review & Proposal

Due: End of Module 8 (Attention & Seq2Seq)

Deliverables:

- ① **Motivation:** Problem statement
- ② **Dataset:** Public URL required
- ③ **Literature Review:**
 - Minimum 3 papers
 - Prior methodologies
 - Identify gaps
- ④ **Proposed Method:** Initial approach
- ⑤ **Experiments:** Evaluation plan

Format Requirements:

- Use arXiv template in Overleaf
- 2-3 pages (excluding references)
- Proper citations
- Clear research questions

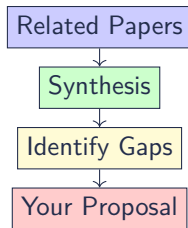
Submission:

- Format:
GroupID_proposal.pdf
- Submit via Canvas

Literature Review Best Practices

What Makes a Good Literature Review?

- **Organized:** Directly related to your research question
- **Synthesized:** Summary of known vs unknown
- **Critical:** Identify controversies and gaps
- **Forward-looking:** Formulate new research questions



Resources

- Google Scholar, arXiv, Papers with Code
- Conference proceedings: NeurIPS, ICML, CVPR, ACL
- Focus on recent work (2020-2025)

Milestone II: Model Architecture & Experiments

Due: End of Module 10 (Vision Transformers)

Key Components

- ① **Abstract:** Concise project summary
- ② **Introduction:** Problem context and motivation
- ③ **Literature Survey:** Expanded from Milestone I
- ④ **Method:** Detailed architecture description
- ⑤ **Preliminary Experiments:** Initial results
- ⑥ **Next Steps:** Planned improvements
- ⑦ **Member Contributions:** Individual responsibilities

Required Elements:

- At least one baseline model
- Training curves & metrics
- Error analysis
- Ablation study plan

Submission Format:

- Compact, up to 3 pages
- Include code repository URL
- Jupyter notebooks welcome
- GroupID_checkpoint.pdf

Experimental Design Guidelines

Baseline Selection:

- Start simple (e.g., ResNet for vision)
- Use pre-trained when applicable
- Document hyperparameters
- Ensure reproducibility

Ablation Studies:

- Vary one component at a time
- Document all changes
- Use consistent seeds
- Track with W&B or TensorBoard

Metrics to Report:

- Task-specific metrics
- Training/validation curves
- Computational efficiency
- Parameter count
- Inference time

Final Project Requirements

Due: After Module 12 (Generative Models)

Deliverables:

- ① **Final Report** (6-8 pages)
 - Full methodology
 - Comprehensive results
 - Thorough analysis
 - Future work
- ② **Code Repository**
 - Clean, documented code
 - README with instructions
 - Requirements file
- ③ **Presentation** (5 minutes)
 - Key findings
 - Live demo if applicable
 - Q&A session

Evaluation Criteria:

- Technical depth
- Innovation
- Experimental rigor
- Result interpretation
- Presentation quality
- Reproducibility

Peer Review:

- Review 2 other projects
- Provide constructive feedback
- Vote for “Best Insight”

Project Ideas by Module Coverage

Computer Vision (Modules 4-5, 10):

- Medical image diagnosis
- Object detection improvements
- ViT vs CNN comparison
- Domain adaptation

NLP (Modules 6-9):

- Text summarization
- Question answering
- Code generation
- Sentiment analysis

Multimodal (Module 11):

- Image captioning
- Visual question answering
- CLIP-style training
- Video understanding

Generative (Module 12):

- VAE for data generation
- GAN improvements
- Diffusion models
- Style transfer

Innovation Opportunities

- Combine architectures. Reproduce advanced SoTA
- Apply models to new domains
- Improve efficiency/speed
- Address ethical considerations

Best Practices for Success

Team Collaboration:

- Use Overleaf for LaTeX
- GitHub for version control
- Regular team meetings
- Clear task division

Experiment Tracking:

- Use W&B or TensorBoard
- Document all hyperparameters
- Save model checkpoints
- Keep experiment logs

Code Quality:

- Modular design
- Clear documentation
- Unit tests for key functions
- Requirements.txt file

Time Management:

- Start literature review early
- Allocate time for debugging
- Plan for compute resources
- Buffer for write-up

Common Pitfalls to Avoid

- Choosing overly complex problems
- Ignoring baseline comparisons
- Poor reproducibility documentation
- Last-minute dataset changes

Resources & Support

Technical Resources:

- UVA Rivanna HPC cluster
- Google Colab (free GPU)
- PyTorch documentation
- D2L textbook examples

Office Hours:

- Dedicated project hours
- TA support sessions
- Ed Discussion forum
- Peer collaboration

Writing Resources:

- Overleaf templates
- arXiv paper format
- Citation management (BibTeX)
- Academic writing guides

Mentorship:

- Primary: Instructor & TAs
- Based on project topic
- Regular check-ins
- Feedback on milestones

Note: This project is your opportunity to create portfolio-worthy work!

Questions & Discussion

Go to the forum!

Next Steps:

- ① Form teams (2-4) members)
- ② Explore dataset options
- ③ Start literature review

Key Takeaways

Public Dataset

! No NDA data!

Follow Timeline

3 milestones

Research Quality

Ablations required

Reproducibility

Clean code

Good luck with your projects!